UPPSALA UNIVERSITET

Data Engineering Project

COVID-19 Dataset Analysis

Alireza Faghihi Moghaddam, Muhammad Hasnain Saeed, Qida Li, Kalle Nordgren, Abhishek Yadav

27 February 2025

1 Introduction

This report investigates the relationship between COVID-19 death rates and vaccination across different states and time periods in the US, as well as the scalability of our data processing framework. In this study, We discussed our experimental setup, presented performance benchmarks under different cluster configurations, and examined correlations between vaccination rates and deaths.

2 Background

The dataset used in this project is sourced from the New York Times, which provides reliable records of COVID-19 cases and deaths across the United States. This dataset has been widely used for epidemiological modeling and public health analysis. We have chosen all the records of COVID-19 deaths from 2020 to 2023 and combined them with Vaccinations in the United States. In this study, we hope to find a relationship between COVID-19 death rates and vaccination rates across different states. By analyzing this data, we aim to identify trends, correlations, and potential causal effects of vaccination campaigns on reducing mortality[1].

Objectives

The questions we aim to answer are directly inspired by [2; 3]. In their study, they have shown the trends of death caused by COVID-19, starting from March 1[2]. They also pointed out that, these deaths occurred before increases in the availability of COVID-19 diagnostic tests and were not counted in official COVID-19 death records[2]. We wanted to expand on this and show The impact of Vaccination on the occurrence of deaths in states and counties since the first submitted record approximately close to the beginning of COVID-19 which was reported around Mid-January to February 2020[4], and when vaccination started. Further, we will show the relationship between vaccination rates and the trends in COVID-19 deaths over time. Our project has two main objectives:

1. **Epidemiological Analysis**: Investigate the relationship between COVID-19 deaths and vaccination rates through the following analyses:
    - **Regional Variations**: Compare deaths-to-cases ratios and vaccination trends across regions to assess disparities.
    - **Correlation Analysis**: Measure the link between full vaccination rates and deaths, both cumulatively and weekly.
    - **Temporal Trends**: Examine how cumulative deaths changed with the start of vaccination programs.
2. **Scalability Analysis**: Evaluate our Spark-based pipeline's efficiency through strong and weak scalability tests.

By addressing these, our study aims to provide a comprehensive picture of how vaccination efforts have influenced COVID-19 mortality trends, while also establishing a robust framework for processing and analyzing large epidemiological datasets.

3 Data format

The datasets are provided in CSV format, which is widely used for tabular data storage and easy integration with data processing tools, especially with Apache Spark (RDDs). CSV was chosen for its simplicity and compatibility, though alternatives like Parquet could offer better performance for large-scale processing. The datasets used in this analysis are structured as follows:

- **COVID-19 Dataset**: The dataset contains about 3,622,862 rows and has columns including: `state`, `county`, `fips`, `cases`, `deaths`, etc.
- **Vaccination Dataset**: The dataset contains about 1,962,781 rows and it provides vaccination statistics by county and state, including: `Administered_Dose1_Recip` etc.

The Tabular format is easy to work with and is compatible with Spark (RDDs), However, Tabular datasets, especially CSV format cant offer hierarchical data types like JSON and their performance in larger dataset sizes is not good. In our project however, we didn't need to change our data format and were able to combine both of these datasets easily, since they were both of the same kind.

4 Computational experiments

4.1 Technologies Used

The project utilized the following technologies:

- **Apache Hadoop (v3.4.1)**: Used specifically for data storage and not for submitting MapReduce jobs.
- **Apache Spark (v3.5.5)**: Configured to enable parallel processing of tasks across multiple nodes, resulting in more efficient use of resources[5].
- **Matplotlib (v3.10.0)**: Soley used for visualizations
- **Pandas (v2.2.3)**: Results from Spark jobs were converted to Pandas dataframe for visualizations.
- **JupyterLab (v4.3.5))**: The programming platform we used for the analysis.

4.2 Experiments Setup

Pseudo-distributed Setup

To better understand the datasets in this project and familiarize ourselves with the configuration and execution of Spark jobs, we set up a single-node pseudo-distributed setup of Apache Hadoop and Spark services. This virtual machine had the following hardware specifications:

Name	vCPUs	RAM	Storage	Services
`group_36_prj`	4	4 GB	20 GB	Master, Worker, NameNode, DataNode, SecondaryNameNode, NodeManager, ResourceManager, SparkSubmit

Fully distributed Setup

To test the scalability of our experiment, we set up a new cluster for our project using three virtual machines with the following hardware specifications:

Name	vCPUs	RAM	Storage	Services
`group_36_prj_cluster_m`	2	4 GB	20 GB	NameNode, Master, Worker, DataNode, SecondaryNameNode
`group_36_prj_cluster_w1`	2	4 GB	20 GB	Worker, DataNode
`group_36_prj_cluster_w2`	2	2 GB	20 GB	Worker

This setup runs a fully distributed Hadoop and Spark which one VM acts as a master node while the other VMs act as Workers. In this setup, HDFS is running with 2 data nodes with a replication factor of 2. Moreover, YARN services were manually stopped as Jobs were run using Spark.

4.3 Experiments

In this project, we analyze COVID-19 case trends and vaccination impacts from both a regional and a state-level perspective. The following subsection details the methodology for each experiment.

Experiment 1: Regional Trend Analysis

- **Datasets Processing**: Both datasets are read from HDFS, and the `date` column is converted into a standard format. Additional fields such as `year`, `month`, and `year_month` are derived for time-based aggregation. A mapping function assigns each state to a corresponding U.S. region, and another maps full state names to abbreviations to ensure consistency between datasets. The data is then aggregated by region and month. For the COVID-19 dataset, total cases, deaths, and the death-to-case ratio are computed. For the vaccination dataset, the total number of first-dose vaccinations is obtained by summing `Administered_Dose1_Recip`.

- **Storage and Visualization**: These aggregated COVID-19 and vaccination results are stored back to HDFS for future reference. Once both aggregated datasets are available, they are converted to Pandas DataFrames for visualization. Two primary visualizations are produced: one displaying the monthly death-to-case ratio by region and another showing the monthly vaccination trends (total first doses administered) by region. These visualizations enable us to compare the temporal progression of the pandemic and the vaccination campaign across different U.S. regions.

Experiment 2: Weekly State-Level Impact Analysis

- **Data Ingestion, Transformation, and Aggregation**: Both datasets are read from HDFS, and the `date` column is converted into a standard format. Additional fields such as `year` and `week` (derived using the `weekofyear` function) are extracted, and records are filtered to include data from 2020 to 2024. A mapping function standardizes state identifiers by converting full state names to abbreviations, ensuring consistency across datasets.

For COVID-19 data, the dataset is aggregated weekly per state by grouping on `state_abbr`, `year`, and `week`. The maximum cumulative cases are computed for each group, and weekly new cases are derived by calculating the difference from the previous week's cumulative count.

For vaccination data, the `Recip_State` column is renamed to `state_abbr` to align with the COVID-19 dataset, and each state is assigned a region using a predefined mapping. The data is aggregated weekly per state, computing the maximum cumulative values for first-dose vaccinations, completed vaccinations, and booster doses. Weekly increments for these vaccination metrics are calculated using window functions.

- **Dataset Merging and Cleaning**: The weekly COVID-19 and vaccination datasets are then merged on `state_abbr`, `year`, and `week`. To address data quality issues, an outlier and anomaly cleaning procedure is applied to key metrics (`new_cases`, `new_dose1`, `new_complete`, and `new_booster`). Values below zero are set to zero, and extreme values exceeding the threshold (of mean plus three standard deviations) are capped.

- **Correlation Analysis**: Finally, a 12-week lag is introduced for the vaccination metrics using window functions, producing lagged columns (`new_dose1_lag`, `new_complete_lag`, and `new_booster_lag`). These lagged values are used to perform correlation analysis with the current week's new cases, providing insights into the delayed impact of vaccination efforts on COVID-19 case trends.

4.4 Scalability Analysis

This section presents our scalability results, focusing on both **strong** and **weak** scalability. Strong scalability measures how execution time changes when increasing the number of cores for the same dataset size, whereas weak scalability examines how well the system handles proportionally larger datasets as more cores become available.

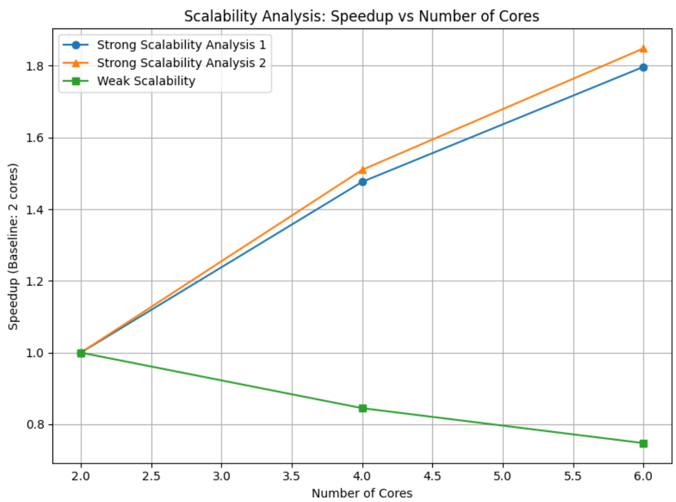

Figure 1: Speedup vs. Number of Cores.

Strong Scalability

For **Experiment 1**, running on 2 cores took 125.88 s, while increasing to 4 cores reduced the time to 85.27 s. Using 6 cores further lowered the execution time to 70.05 s. We see a clear improvement in performance as cores increase, though the speedup is not perfectly linear. This suggests that while more cores help reduce overhead and parallelize data transformations, there are still factors (like shuffling and scheduling) that limit perfect scaling.

For **Experiment 2**, using 2 cores resulted in 105.59 s, 4 cores cut it down to 69.94 s, and 6 cores reduced the time to 57.12 s. Here, too, the overall trend indicates that more cores significantly decrease execution time, though again not in direct proportion to the number of cores.

Weak Scalability

This scalability analysis was conducted on **Experiment 1** script. To test weak scalability, we proportionally increased both the dataset size and the number of cores. The original dataset was duplicated 2x for the 4-core run and 3x for the 6-core run. When processing the base dataset on 2 cores, the execution time was 68.21 s. Doubling the dataset size (2x) and using 4 cores resulted in 80.71 s while tripling the dataset size (3x) on 6 cores gave 91.23 s. Although performance degrades somewhat as the dataset size grows, the rise in execution time remains moderate. This suggests that our Spark-based solution can handle bigger inputs reasonably well as more cores are allocated, though overheads such as shuffling, scheduling, and I/O become more significant when the data volume increases.

Discussion of Observed Scalability

Overall, these results (Figure 1) show that adding more cores yields faster execution in both analyses, confirming a beneficial parallelization effect. However, the speedup is not linear, showing that overheads and potential bottlenecks (e.g., shuffle operations) limit perfect scaling. For weak scalability, while execution time does rise with bigger datasets, the increase is moderate enough to show that our Spark-based approach can manage larger inputs if sufficient resources are allocated.

4.5 Results and Insights

Experiment 1: Regional Trend Analysis

Figure 2 shows the monthly deaths-to-cases ratio by region, while Figure 3 presents the monthly vaccination trends. We can see that the death-to-cases ratio initially spikes—likely due to limited testing and higher early mortality—but gradually declines as testing and treatments improve across all regions. Meanwhile, vaccination uptake starts near zero and steadily increases, with some regions showing faster growth. As first-dose vaccinations climbed, the death-to-cases ratio stabilized or continued trending downward, suggesting that higher vaccination coverage contributed to reducing severe outcomes.

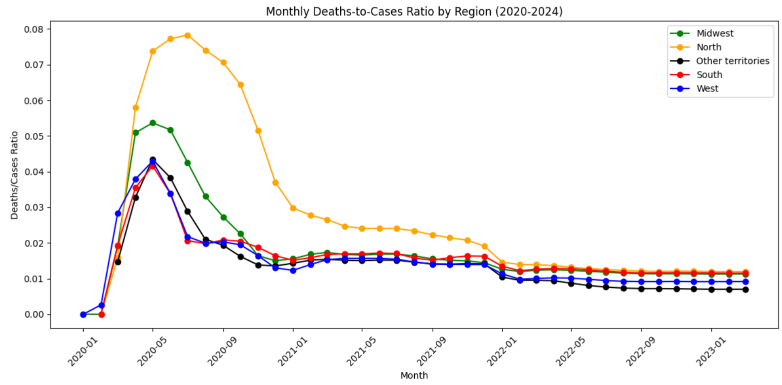

Figure 2: Monthly Death-to-Cases Ratio by Region (2020–2024)

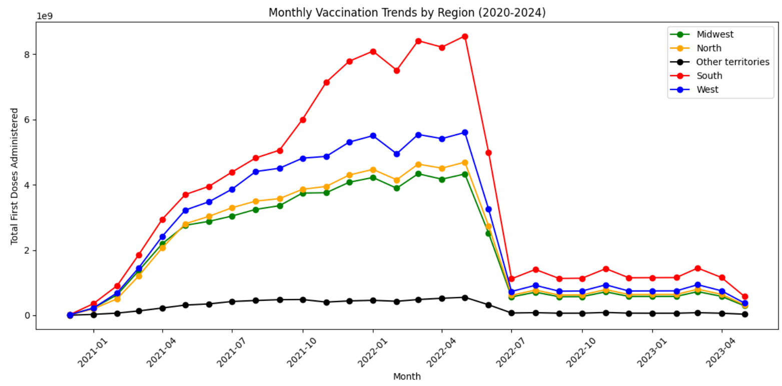

Figure 3: Monthly First Doses Administered by Region (2020-2024)

Experiment 2: Weekly State-Level Impact Analysis

Metric	Correlation with New Cases
new_dose1	0.559
new_complete	0.483
new_booster	0.406
new_dose1_lag	0.230
new_complete_lag	0.281
new_booster_lag	0.142

Table 1: Correlation between weekly new cases and vaccination metrics

The results in Table 1 show that when more vaccinations are administered in a week, there is a stronger association with higher new cases during that same week—possibly because vaccination campaigns are often reactive to case surges. On the other hand, the lower correlations observed for the vaccination data from 12 weeks earlier suggest that the effect of vaccinations on reducing cases is not immediately clear. In simple terms, while immediate vaccinations are closely linked with current case surges, the impact of vaccines measured 12 weeks later is weaker, likely because it takes time for vaccines to build immunity. It is important to note that these correlations do not prove causation; they only indicate how closely the numbers move together, and many other factors could be influencing these relationships given the simplicity of the analysis.

5 More Insights

Trends of COVID-19 Cases from 2020-2023

Our COVID-19 dataset holds the record of deaths, starting from January 1st, 2020 to 31 December 2020. During this period, we noticed that New York City had the highest number of reported cases of death caused by COVID, with a total number of 25144. It can be seen from the table, that our findings suggest while this trend continued, the slope decreased over time.

Table 2: Cumulative COVID-19 Deaths in New York City by Year

Year	Cumulative Deaths
2020	25144
2021	35382
2022	43935
2023	45155

Year	**1 Worker, 2 Cores**	**2 Workers, 2 Cores**
2020	6.3129	8.2326
2021	5.0530	4.8446
2022	6.4416	4.7897
2023	5.1466	1.6261

Table 3: Performance Metrics for Different Worker and Core Configurations

Correlation between Vaccination rate and deaths

In our dataset, death counts are not only recorded but also continuously updated for each county and state over time. This means each time a death is recorded, it gets added to the total amount of that county. Having that in mind, we calculated the correlation between full vaccination rates and deaths to see if higher vaccination rates are linked to fewer deaths. Normally a negative correlation suggests that we have fewer deaths with vaccination, but since the numbers on our dataset are added to the previous record, not changed into it, we predict that the correlation should at best be near zero, which was the case. This suggests that by the time vaccination was completed, the rate of increase in deaths had slowed, and the increase reached zero[1].

corr(Series_Complete_Pop_Pct, deaths)
0.1575377800186528

Table 4: Correlation between full vaccination rate and deaths

6 Discussion and Conclusion

Our analysis of COVID-19 vaccination and mortality trends provides key insights into the relationship between vaccination rates, case surges, and regional disparities. While a direct negative correlation between full vaccination rates and deaths was not evident, our findings suggest that higher vaccination coverage slowed the rate of increase in deaths over time. Additionally, we observed a strong positive correlation between weekly new vaccinations and new cases, likely due to vaccination efforts being reactive to surges in infections.

Geographically, the highest number of COVID-19 deaths were concentrated in densely populated urban regions, particularly in the northern U.S. In contrast, lower death rates were recorded in less populated rural areas, such as the Midwest and Mountain West. Furthermore, higher vaccination rates were observed in the South and West Coast, where early vaccine distribution and public health campaigns were more effective.

From a technical perspective, our scalability analysis demonstrated that Apache Spark efficiently processed large datasets, leveraging parallel computing. However, we observed diminishing returns when increasing the number of cores, likely due to overhead from shuffle operations and job scheduling inefficiencies. While weak scalability results showed moderate increases in execution time as dataset size grew, our framework remained capable of handling larger inputs effectively.

References

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